Answer key (Module 20 exercises)

```spl
| tstats summariesonly=true count FROM datamodel=Authentication
  WHERE Authentication.action="failure" BY Authentication.src
| rename Authentication.* AS *
| sort - count
```

```ini
[monitor:///var/log/syslog]
index = linux_sys
sourcetype = syslog
disabled = 0
```

```bash
sudo /opt/splunk/bin/splunk btool props list linux_secure --debug | grep TRUNCATE
```

Ex. 4, first four checks for `index=windows` empty:

1. Does the `windows` index exist **on the indexer**, and is your role allowed to search it?
2. Is the UF running, and does its service account have permission to read the Security log?
3. Is the UF connected to the indexer (`splunk list forward-server`, receiver port 9997)?
4. Search `index=_internal source=*splunkd.log* (ERROR OR WARN)` for the host, and set the time range to All time.

Self-check key (Module 20):

1. A data model defines the data, acceleration pre-builds summaries, and `tstats` reads those summaries quickly.
2. sourcetype → extractions, aliases and calculated fields → eventtype → tag → data model constraint.
3. `app/local` wins, because `local` always beats `default`, and `default` is overwritten on upgrade.
4. `--debug` shows which file each setting came from.
5. In `inputs.conf` at the input, or in `props.conf` rules.

📘 MODULE 27: DASHBOARDS

27.1 Concept

A **dashboard** is a page of **panels**. Each panel runs a search and displays the result as a table, chart or single value. Dashboards can have **inputs** (time picker, dropdowns) that feed **tokens** into the searches.

27.2 Why it matters

L1 analysts **read** dashboards all day, and L2 analysts **build** them. Interviewers often ask "walk me through a dashboard you built."

27.3 Types

Type	Format	Use
Classic dashboard	Simple XML	Mature, simple, and very common in companies
Dashboard Studio	JSON, drag-and-drop canvas	Newer, more visual. Some Simple XML features differ.

This module uses **classic Simple XML** because it is the most widely used. Verify Dashboard Studio feature parity in current docs.

27.4 Building a panel from a search (fastest way)

1. Run a search, for example `index=_internal sourcetype=splunkd | timechart span=1h count by log_level`.
2. Open the **Visualization** tab and choose a chart type.
3. Click **Save As** → **New Dashboard** (or **Existing Dashboard**).
4. Name it, choose permissions (Private or Shared in App), and **Save**.
5. Open it and click **Edit** to change layout, titles, and inputs.

27.5 Panel types

Panel	Best for
Table	Details (IPs, users, counts)
Column/Bar	Comparing categories (top 10 IPs)
Line/Area	Trends over time (`timechart`)
Single value	One KPI (failed logins today)
Pie	Few categories only

27.6 Time ranges and tokens

- **Time input** gives `$tr.earliest$` and `$tr.latest$` tokens for every panel.
- **Dropdown input** gives `$host_tok$` for use in the search text.
- **Drilldown** sets a token when a user clicks a row, and a hidden panel appears with details for that value.
- Token rule: use `$name$` in searches. Put `*` as the default value so "All" works.

27.7 Dashboard 1: Login Monitoring (full Simple XML)

Create via **Dashboards** → **Create New Dashboard** → **Classic Dashboards**, then **Source** to paste. Adjust `index`, `sourcetype` and the `rex` to your data.

xml

```
<dashboard version="1.1">
  <label>Login Monitoring</label>
  <fieldset submitButton="false">
    <input type="time" token="tr">
      <label>Time range</label>
      <default>
        <earliest>-24h@h</earliest>
        <latest>now</latest>
      </default>
    </input>
    <input type="dropdown" token="host_tok" searchWhenChanged="true">
      <label>Host</label>
      <choice value="*">All</choice>
      <default>*</default>
      <fieldForLabel>host</fieldForLabel>
      <fieldForValue>host</fieldForValue>
      <search>
        <query>| tstats count WHERE index=linux_auth BY host</query>
        <earliest>-24h</earliest>
        <latest>now</latest>
      </search>
    </input>
  </fieldset>
  <row>
    <panel>
      <title>Failed logins over time</title>
      <chart>
        <search>
          <query>index=linux_auth sourcetype=linux_secure "Failed password" host=$host_tok
| timechart span=1h count</query>
          <earliest>$tr.earliest$</earliest>
          <latest>$tr.latest$</latest>
        </search>
        <option name="charting.chart">column</option>
      </chart>
    </panel>
  </row>
  <row>
    <panel>
      <title>Top source IPs (click a row)</title>
      <table>
        <search>
          <query>index=linux_auth sourcetype=linux_secure "Failed password" host=$host_tok
| rex "from (?&lt;src_ip&gt;\d{1,3}(?:\.\d{1,3}){3})"
| stats count, dc(user) AS users_tried by src_ip
```

```
| sort - count
| head 10</query>
          <earliest>$tr.earliest$</earliest>
          <latest>$tr.latest$</latest>
        </search>
        <drilldown>
          <set token="ip_tok">$row.src_ip$</set>
        </drilldown>
      </table>
    </panel>
    <panel depends="$ip_tok$">
      <title>Events for $ip_tok$</title>
      <table>
        <search>
          <query>index=linux_auth sourcetype=linux_secure "$ip_tok$" | table _time host _r
          <earliest>$tr.earliest$</earliest>
          <latest>$tr.latest$</latest>
        </search>
      </table>
    </panel>
  </row>
</dashboard>
```

Key points:

- Inside XML, `<` and `>` must be written `<` and `>`, which is why the regex looks different. Forgetting this is the most common error.
- `depends="$ip_tok$"` hides the second panel until a row is clicked.
- `$row.src_ip$` is the clicked row's value.

27.8 Dashboard 2: Web Log Dashboard (panel searches)

(Assumes Apache data with `sourcetype=access_combined`.)

spl

```
index=web sourcetype=access_combined | timechart span=1h count by status
index=web sourcetype=access_combined | top limit=10 clientip
index=web sourcetype=access_combined status>=400 | stats count by uri_path | sort - count
index=web sourcetype=access_combined | stats sum(bytes) AS total | eval MB=round(total/10
```

Use the last one as a **single value** panel.

27.9 Dashboard 3: System Log Dashboard

spl

```
index=linux_sys sourcetype=syslog | timechart span=1h count by host
index=linux_sys sourcetype=syslog (error OR fail*) | stats count by host | sort - count
```

27.10 Dashboard 4: Splunk Health Dashboard

```spl
index=_internal source=*splunkd.log* log_level=ERROR | timechart span=1h count by componer
| tstats count WHERE index=* BY index, _time span=1h
index=_internal source=*license_usage.log type=Usage | timechart span=1d sum(b) AS bytes
| tstats latest(_time) AS last_seen WHERE index=* BY host | eval mins=round((now()-last_s
```

The license search works on the license manager (your standalone lab qualifies). The second line is tstats volume by index and time, and the last is the silent-host panel.

27.11 Performance tips for dashboards

- Use **tstats** or **saved-search-based panels** for heavy panels.
- Share a **base search** and let panels **post-process** it, so one search feeds many panels.
- Avoid `index=*` over All time.
- Use a sensible default time range (24h, not All time).
- Too many auto-refreshing real-time panels hurt performance.

27.12 Hands-on

1. Build Dashboard 1 with your `linux_auth` data (or adapt it to `_internal`).
2. Add a single-value panel: failed logins in the last 24 hours.
3. Add a dropdown of `sourcetype` that filters one panel.
4. Test the drilldown by clicking a row.

27.13 Troubleshooting

Symptom	Cause
Panel says “Search did not return any results”	Token not set or wrong default, time range empty, wrong index
“Waiting for input”	Token never set, so add a default value
Panel spinner forever	Heavy search, so use tstats/accelerate/narrow time
XML error	`<` / `>` / `&` not escaped
Works for me, not others	Dashboard, saved search, or lookup permissions are private
Drilldown does nothing	Token name mismatch, or `$row.field$` doesn’t exist in the table

27.14 Interview Q&A

- **What is a token?** A variable set by inputs or drilldown, and substituted into searches as `$name$`.
- **How do you speed up a dashboard?** Base search with post-process, `tstats`, accelerated data models, narrower time, scheduled searches.
- **Classic vs Dashboard Studio?** Classic uses Simple XML, Studio uses JSON with a free-form canvas.
- **How do you build a drilldown?** `<drilldown><set token="...">$row.field$</set>` and use the token in another panel.

27.15 Revision

Panels run searches · tokens `$x$` · time input `$tr.earliest$` · escape `< >` · base search + post-process · share permissions.